

PHYSICS FORMULA CHEATSHEET

Secondary School · SI units: m, kg, s, N, J, W, Pa, A, V, Ω, Hz

1. Motion (Kinematics)

$$v = \frac{s}{t} \quad \text{speed}$$
$$a = \frac{v - u}{t} \quad \text{acceleration}$$
$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$
$$s = \frac{(u + v)}{2} t$$

u = initial vel., v = final vel., $g \approx 9.8 \text{ m/s}^2$ (free fall)

2. Forces & Newton's Laws

$$F = ma \quad \text{Newton's 2nd law}$$
$$W = mg \quad \text{weight}$$
$$p = mv \quad \text{momentum}$$
$$F = \frac{\Delta p}{t} \quad \text{force as rate of change of momentum}$$
$$F_f = \mu N \quad \text{friction}$$
$$F = kx \quad \text{Hooke's law (spring)}$$
$$F = \frac{Gm_1m_2}{r^2} \quad \text{gravitation}$$

Conservation of momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

3. Work, Energy & Power

$$W = Fs \cos \theta \quad \text{work done}$$
$$E_k = \frac{1}{2}mv^2 \quad \text{kinetic energy}$$
$$E_p = mgh \quad \text{gravitational PE}$$
$$E_s = \frac{1}{2}kx^2 \quad \text{elastic PE}$$
$$P = \frac{W}{t} = Fv \quad \text{power}$$
$$\eta = \frac{\text{useful output}}{\text{total input}} \times 100\% \quad \text{efficiency}$$

4. Circular Motion & Moments

$$v = \frac{2\pi r}{T} \quad \text{orbital speed}$$
$$a_c = \frac{v^2}{r} = \omega^2 r \quad \text{centripetal acc.}$$
$$F_c = \frac{mv^2}{r} \quad \text{centripetal force}$$
$$\omega = \frac{2\pi}{T} = 2\pi f \quad \text{angular velocity}$$
$$M = Fd \quad \text{moment (torque)}$$

Equilibrium: $\sum M_{cw} = \sum M_{acw}$

5. Pressure & Density

$$\rho = \frac{m}{V} \quad \text{density}$$
$$P = \frac{F}{A} \quad \text{pressure}$$
$$P = \rho gh \quad \text{pressure in a liquid}$$
$$F_b = \rho_f Vg \quad \text{upthrust (Archimedes)}$$
$$P_1V_1 = P_2V_2 \quad \text{Boyle's law (const. } T\text{)}$$

6. Heat & Thermal Physics

$$Q = mc\Delta T \quad \text{specific heat capacity}$$
$$Q = mL \quad \text{latent heat (change of state)}$$
$$T(\text{K}) = \theta(^{\circ}\text{C}) + 273 \quad \text{Kelvin scale}$$
$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \text{pressure law (} V \text{ const.)}$$
$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{Charles' law (} P \text{ const.)}$$
$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad \text{combined gas law}$$

7. Waves, Sound & Light

$$v = f\lambda \quad \text{wave equation}$$
$$f = \frac{1}{T} \quad \text{frequency}$$
$$n = \frac{\sin i}{\sin r} = \frac{c}{v} \quad \text{refractive index}$$
$$\sin C = \frac{1}{n} \quad \text{critical angle}$$
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad \text{lens / mirror eqn.}$$
$$m = \frac{v}{u} = \frac{h_i}{h_o} \quad \text{magnification}$$
$$v_{\text{sound}} = \frac{2d}{t} \quad \text{echo}$$

$c = 3.0 \times 10^8 \text{ m/s}$ (light), sound in air $\approx 340 \text{ m/s}$. Law of reflection: $i = r$.

8. Electricity

$$I = \frac{Q}{t} \quad \text{current}$$
$$V = \frac{W}{Q} \quad \text{potential difference}$$
$$V = IR \quad \text{Ohm's law}$$
$$R = \frac{\rho L}{A} \quad \text{resistance of a wire}$$
$$P = IV = I^2R = \frac{V^2}{R} \quad \text{electrical power}$$
$$E = Pt = IVt \quad \text{electrical energy}$$
$$R_s = R_1 + R_2 + \dots \quad \text{series}$$
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \quad \text{parallel}$$
$$\mathcal{E} = V + Ir \quad \text{emf \& internal resistance}$$

Series: same I , V splits. Parallel: same V , I splits.

9. Magnetism & Electromagnetism

$$F = BIL \quad \text{force on a current-carrying wire}$$
$$F = Bqv \quad \text{force on a moving charge}$$
$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \quad \text{transformer}$$
$$V_p I_p = V_s I_s \quad \text{ideal transformer (power)}$$

Fleming's left-hand rule (motor), right-hand rule (generator).

10. Atomic & Nuclear Physics

$$A = Z + N \quad \text{mass no. = protons + neutrons}$$
$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}} \quad \text{radioactive decay}$$
$$E = hf \quad \text{photon energy}$$
$$E = mc^2 \quad \text{mass-energy equivalence}$$

α : ${}^4_2\text{He}$ β^- : electron γ : EM wave.
After n half-lives, fraction left = $(\frac{1}{2})^n$.

11. Useful Constants

g	9.8 m/s^2	c	$3.0 \times 10^8 \text{ m/s}$
G	$6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$	h	$6.63 \times 10^{-34} \text{ J s}$
e	$1.6 \times 10^{-19} \text{ C}$	P_{atm}	$1.01 \times 10^5 \text{ Pa}$
ρ_{water}	1000 kg/m^3	c_{water}	4200 J/kg K

12. Unit Prefixes & Tips

G	10^9	M	10^6	k	10^3
c	10^{-2}	m	10^{-3}	μ	10^{-6}

- Convert to SI units *before* substituting.
- Write the formula, substitute, then state the answer with units.
- Check: speed in km/h $\div 3.6 \rightarrow$ m/s.